

An “Image Theory” of RPM*

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Abstract

We show how a brand manufacturer’s control over retail prices coordinates a channel of competing retailers when consumers rely on retail prices as an indicator of product quality. We first establish how higher retail prices can be associated with both higher perceived quality and higher actual quality. We next identify two distinct conflicts of interest between retailers and the manufacturer. Individual retailers do not internalize the cross-outlet reputation spillover that their price choices have on overall brand image. They also have weaker incentives to support brand image through higher prices, as elevated quality perceptions strengthen the manufacturer’s position in subsequent contract negotiations. Our channel-coordination rationale for resale price maintenance (RPM) thus applies in retail settings where retailers need not be incentivized to undertake non-contractible service activities, as the key opportunism problem with respect to quality provision lies between the manufacturer and consumers.

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1 Introduction

For many fast-moving consumer goods, notably in food retailing, quality depends on a manufacturer’s ongoing effort, such as procuring high-quality inputs or securing high hygienic standards in production and handling. Consumers, in turn, may not always be able to assess a product’s quality before purchasing and may, among other things, rely on retail prices as an indicator of quality. In this article, we first set up a simple model that motivates a positive relationship between prices and both perceived and actual quality. Next, we show how this relationship gives rise to a conflict of interest between retailers and manufacturers, as manufacturers prefer a strictly higher price. When prices are set in retailers’ individual interests, they do not internalize the full reputation spill-over for the manufacturer’s good. The conflict of interest is further aggravated by the fact that prices are also chosen to influence retailers’ and manufacturers’ bargaining positions, which depend on consumers’ perceptions of the product’s quality. Overall, equilibrium quality and efficiency can be higher when manufacturers exert greater control over retail prices through retail price maintenance (RPM).

Our contribution to the literature on vertical channels and RPM is to identify a *channel-coordination* rationale for manufacturer control over retail prices that does not rely on non-contractible, sales-enhancing activities of retailers. The rationale applies in particular to branded fast-moving consumer goods, such as in food retailing or cosmetics, where even the staff of “bricks and mortar” retailers does not interact directly with consumers, e.g., to provide advice. Our theory thus provides a rationale for why branded-goods manufacturers may be particularly eager to keep control over the retail price even when retailers need not be incentivized to provide such services and even when, as we show, intrabrand competition is not the main concern. The driver, in our model, is the link between price and quality image together with the described conflicts of interest between retailers and the concerned manufacturer.

The mechanism we identify is distinct from the dominant rationales in the channels and RPM literature, which build on retailers’ non-contractible actions: provision of services and demand-enhancing activities (Telser 1960; Klein and Murphy 1988; Mathewson and Winter 1998), inventory provision under demand uncertainty (Deneckere et al. 1997), or quality certification through retailer assortment choice (Marvel and McCafferty 1984). In our setting, retailers provide no such services; the opportunism problem instead lies between the manufacturer and consumers, and RPM coordinates competing retailers around a

price level that supports the brand’s quality image. The empirical premise on which our analysis rests — that consumers infer product quality from observed retail prices — is well established in the marketing literature: the meta-analytic review by Völckner and Hofmann (2007) documents a robust positive association between price and both perceived and actual quality across product categories (cf. also Rao 1984; Erickson and Johansson 1985; Leavitt 1954).

In this respect, our channel-coordination rationale for RPM may also apply when various distribution channels, such as “offline” and “online,” do not differ in the provision of other services to consumers, such as advice or shopping experience. Hence, in these cases, there may not be an efficiency defense based on the provision of incentives or, likewise, the protection of such incentives in one (“offline”) channel against competition from another (“online”) channel where such services are not provided. Attempts by brand manufacturers to control prices in online distribution have, in fact, been at the forefront of various antitrust cases.¹ The European Union’s block exemption regulation for vertical restraints and the respective guidelines consequently make specific reference to online sales. While the text recognizes the role of a product’s image as a justification for selective distribution systems, this does not extend to controlling prices across channels to uphold such an image.² Our theory suggests that this approach is too narrow to account for the relevance of a product’s image.

An “image theory” of RPM has also been occasionally acknowledged in the legal lit-

¹For instance, Lindsay (2011) reports, in particular, two recent cases of, inter alia, minimum RPM provisions for online sales (Bioelements, a skin-care products manufacturer, and Tempur-Pedic, a seller of premium mattresses). He observes a substantially different treatment of these clauses by the respective Californian and New York Attorneys General, suggesting that also in the US, the discussion about RPM is far from over even after the much discussed Leegin decision of the Supreme Court in 2007.

²Also, a differential treatment of online distribution channels seems to be justified only when these lack the provision of additional services, thus mirroring theories that, in contrast to ours, build on retailer moral hazard. The Vertical Restraints Block Exemption Regulation and accompanying Guidelines came into force on 1 June 2010. For instance, with regard to online distributors, manufacturers may require the provision of after sales services or the presence of physical locations and showrooms. Cf. European Commission Regulation (EU) No 330/2010 of 20 April 2010 on the application of Article 101(3) of the Treaty on the Functioning of the European Union to categories of vertical agreements and concerted practices, OJ L 102, 23.4.2010, and European Commission Notice - Guidelines on vertical restraints, Official Journal C 130, 19.05.2010, p. 1.

erature,³ though to our knowledge it has been largely ignored by competition economics.⁴ Notable exceptions are Bontems and Mahenc (2014), who consider price signaling in vertical relationships where manufacturers and retailers are better informed than consumers. They emphasize the role of vertical contracts in solving the multiplicity of signaling equilibria. Dertwinkel-Kalt and Köster (2022) provide a rationale for “brand protection” in a model of salience-driven attention. They show that price disparities across channels (e.g., online discounts) attract disproportionate consumer attention to prices, which mechanically lowers the appreciation for quality. In this view, RPM protects brand image not by signaling quality, but by preventing a “price-salient” environment that erodes perceived value.

In our model, a higher price is associated with higher quality and a higher perception of quality by those consumers who do not directly observe the quality choice. When the manufacturer controls the retail price, he will fully internalize the benefits from an overall higher perception of quality. Precisely, when consumers form beliefs based on their perception of prices across all retailers and shopping trips, the manufacturer internalizes this. In contrast, retailers tend to free-ride on the quality perception generated by other retailers’ choice of prices. We derive conditions for when an induced increase in quality dominates the deadweight loss from a higher price, which is the case precisely when quality perceptions matter more, as more consumers must rely on them for their purchases.

A second channel through which such a difference in interests between manufacturers and retailers affects prices and quality is the effect that the product’s (perceived) quality has on the outside options of manufacturers and retailers. Retailers may set a lower price also to induce lower quality perceptions, which decreases the outside option of the manufacturer and enhances retailers’ own outside option in case they later stock a different product. Notably, under our chosen bargaining solution, such a conflict of interest would *not* arise when quality was exogenous, as then retailers and the manufacturer would choose the same price. Hence, the fact that quality is endogenous and affected by prices is essential

³Among legal scholars, for instance, Orbach (2008, 2010) builds a legal defense of RPM on such an “image theory” of prices. The argument there rests however on a more diffuse notion of “brand image,” which is affected by price choice and which is assumed to generate consumer welfare. It relates to Leibenstein’s (1950) theory of “conspicuous consumption” and consumers’ willingness to pay a price “above the intrinsic value” to achieve exclusivity. (This goes directly back to Veblen (1899), as well as Taussig’s (1916) notion of “articles of prestige.”) As we equate “brand image” with “quality image” in our model, we can however derive potential efficiency gains directly from more standard consumer preferences.

⁴In Marvel and McCafferty (1984) low-cost retailers may free-ride on the quality certification provided by higher-cost retailers. While a higher margin promised to retailers thus ensures that also high-cost retailers stock (and thereby certify) a high-quality good, it is the availability of products at these retailers and not a particular price itself that generates the respective quality perception.

for this potential conflict of interest between retailers and manufacturers to arise in our model. Again, quality is higher when the manufacturer controls the price.

In our baseline model, a higher retail price induces higher equilibrium quality whenever the manufacturer’s quality choice is observed by at least some consumers — or, equivalently in our specification, observed with positive probability by any individual consumer. The remaining consumers do not directly observe quality and form rational expectations about it given the prices they see. This model and its analysis serve two purposes. First, to some extent, it is of interest in its own right, as we clearly isolate, already in this simple model, three channels for why consumers may rightly anticipate a positive relationship between prices and quality. Second and more importantly, the model provides a simple starting point to introduce the conflict of interest between retailers and the considered manufacturer. That said, we are aware that a number of theoretical papers in the industrial organization literature have associated price and quality perception, albeit not in the context of vertical relationships and not applied to antitrust issues, notably Klein and Leffler (1981), Shapiro (1983), and Wolinsky (1983).⁵

To model negotiations between a manufacturer and several possibly competing retailers, we use a bargaining approach that extends an idea from Inderst and Wey (2003). Under this solution concept, each bilateral contract specifies a “fair” sharing rule for the net surplus generated under all possible agreements (coalitions). As we will discuss, the obtained solution is closely related to other recent solution concepts for multilateral negotiations, such as De Fontenay and Gans (2014). One of our contributions is to show the applicability of this solution concept, as in our setting it gives rise to relatively simple and intuitive expressions.

Romano (1994) is closer to our setting in that both manufacturer and retailers make non-price decisions affecting demand, but it does not consider price-image as the link between retail prices and the manufacturer’s quality choice. For branded products in areas such as grocery or cosmetics, where retailers typically do not provide advice or other consumer-facing services, the key opportunism problem lies between manufacturers and

⁵Also, in models where quality is exogenously given, high-quality firms may use high prices to signal the superiority of their products (e.g., Bagwell and Riordan (1991) or Milgrom and Roberts (1986), where high prices are used in combination with dissipative advertising). Noteworthy may also be a very early article in the *Review of Economic Studies* by Scitovszky (1944), who notes: “[...] in the United States ‘expensive’ is in the process of losing its original meaning and becoming a synonym for superior quality. Worse still, one of the largest American breweries uses the advertising slogan: ‘Michelob, America’s highest-priced beer!’” According to his reasoning and in line with the subsequent formalizations, including the one given in this paper, uninformed consumers can rely on a positive relationship between price and demand, as “expert consumers” would not buy a high-price product if it was of low quality.

consumers, and our channel-coordination mechanism applies directly.

The rest of this paper is organized as follows. Section 2 introduces the baseline model and establishes a relationship between price and quality. Section 3 introduces retailers. Section 4 considers non-competing retailers and isolates a conflict of interest that arises from free-riding on quality (brand) image. In Section 5 we introduce retailer competition. Section 6 offers some concluding remarks.

2 The Relationship between Price and Quality

The mechanism at the heart of this paper is that some consumers infer product quality from observed retail prices. We treat this price-quality inference as a reduced-form behavioral premise grounded in the marketing evidence summarized in Völckner and Hofmann (2007), and model it compactly through a fraction $1 - \gamma$ of consumers who do not directly observe quality and form rational expectations about it given the prices they observe. This specification is consistent with the classical signaling literature in which prices convey quality information (Bagwell and Riordan 1991; Wolinsky 1983; Klein and Leffler 1981; Shapiro 1983), but the mechanism we isolate is conceptually distinct. In separating-signaling models, equilibrium price distortions are *larger* when more consumers are uninformed, because a high price must deter low-quality imitation. In our model, by contrast, the manufacturer's preferred price moves in the *opposite* direction: it rises with the share of informed consumers γ because the source of distortion is the manufacturer's own ex-post incentive to skimp on quality once retail prices are set, not a deterrence problem against imitating types. The contribution of the paper is therefore a commitment-and-coordination story within a vertical channel, not a signaling story.

2.1 Model

In this section, we first consider an auxiliary model to introduce the basic relationship between price and quality. In this model, a manufacturer sells directly to final consumers; thus, we presently abstract from the presence of intermediaries. Demand for the firm's product depends on the product's price p and a scalar indicator of quality $q \geq 0$: $D(p, q)$. The supplier's costs depend on quantity $x = D(p, q)$ and quality: $c(x, q)$. As we will discuss in what follows, the positive relationship between price and quality that we derive in this section holds generally whenever $c_{qx} \geq 0$, i.e., when per-unit costs of production are

weakly increasing in quality.⁶ Our motivational examples in the introduction, however, suggest a specification where the costs of ensuring higher quality change proportionally with quantity: $c(x, q) = k(q)x$. For instance, this should be the case when higher costs result from the procurement of higher-quality inputs or from ensuring higher hygienic and safety standards in production and shipment. Here, $k(q)$ is assumed to be a twice differentiable function with $k'(q) > 0$ for $q > 0$ and $k''(q) \geq 0$. The supplier's profits are $\Pi = D(p, q)[p - k(q)]$. We further specify that $D_p < 0$ and $D_q > 0$ where $D > 0$.

For the present auxiliary analysis, we now consider the following game. We subsequently discuss the particular timing that we thereby choose. First, in $t = 1$, the firm chooses a price p . Then, in $t = 2$, it chooses a quality q . Finally, in $t = 3$, consumers decide whether to purchase or not. We discuss the sequence of timing in $t = 1$ and $t = 2$ below. Furthermore, before purchasing, a consumer only observes the true quality with probability γ . As we restrict consideration to equilibria in pure strategies, we denote an uninformed consumer's beliefs by $\hat{q}$. These will depend on the observed price choice.

Before analyzing this game, we comment on the choice of the sequence of timing in $t = 1$ and $t = 2$. Our focus throughout this paper is on price as a (relatively) longer term choice (cf. also the additional discussion in Section 3). It is part of the overall positioning of the product, i.e., its branding. In practice, the decision on the overall price level must then be complementary to the other marketing choices, such as the scope and content of the advertising campaign. While surely some key (quality) features of the product are also chosen for the long term, particularly concerning fast moving consumer goods and notably branded food products, we consider decisions that must be made constantly to maintain high quality. As mentioned previously, this could concern the procurement of high-quality inputs or the overall conditions of production and handling of the product (e.g., with regard to food safety). The manufacturer could be tempted to save costs by reducing care or using cheaper inputs. That said, all that is needed for what follows is that there is a positive relationship between price (or, anticipating the subsequent extension to vertical relationships, a manufacturer's expected margin) and both perceived as well as actually chosen quality (which are then the same in equilibrium). In particular, building on the literature mentioned in the Introduction, we could also have reversed the timing, so that price is chosen before quality, which would then give rise to a game of signaling (and the need to then refine the set of equilibria).

⁶This is, however, not a necessary condition but only a sufficient one.

2.2 Equilibrium of the Auxiliary Game

We solve the game backwards. Consumers' decisions in $t = 3$ are already captured by the demand function. A given consumer is only informed with probability γ about the true quality choice. With probability $1 - \gamma$, a consumer is uninformed and has the beliefs $\hat{q}$. Demand is thus given by

$$\hat{x} = \gamma D(p, q) + (1 - \gamma) D(p, \hat{q}),$$

so that firm profits are

$$\pi(p, q, \hat{q}) = \hat{x} [p - k(q)], \quad (1)$$

depending on price, actual quality, and quality beliefs. Actual quality affects demand only when the consumer is informed. For given beliefs $\hat{q}$, the optimal quality q_{BR} is thus determined from (1) by the following first order condition:

$$\left. \frac{d}{dq} \pi(p, q, \hat{q}) \right|_{q=q_{BR}} = \gamma [p - k(q_{BR})] D_q(p, q_{BR}) - \hat{x}_{BR} k'(q_{BR}) = 0, \quad (2)$$

where we used $\hat{x}_{BR} = \gamma D(p, q_{BR}) + (1 - \gamma) D(p, \hat{q})$. (Here, the notation q_{BR} refers to the fact that this is the “best response” to a particular choice of the price and consumers' beliefs.) We stipulate that π is strictly concave in q so that there is always a unique value q_{BR} . This, as well as uniqueness for the subsequently solved programs, holds, in particular, for the (linear-quadratic) specification that is used for illustration below.

In equilibrium, given the price p that is set initially and is observed by all consumers, beliefs must be rational. In a slight extension of notation, it is thus required that $\hat{q} = q_{BR}(\hat{q})$ (where it is convenient to suppress the dependency on the price for now). We denote this level, for given p , by $\hat{q}^*$. Using the condition for q_{BR} from (2), $\hat{q}^*$ must solve

$$z(p, \hat{q}^*) := \gamma [p - k(\hat{q}^*)] D_q(p, \hat{q}^*) - D(p, \hat{q}^*) k'(\hat{q}^*) = 0. \quad (3)$$

Again, we assume that this gives rise to a unique interior solution and obtain from implicit differentiation

$$\frac{d\hat{q}^*}{dp} = \frac{1}{-z_{\hat{q}^*}} [\gamma D_q(p, \hat{q}^*) - k'(\hat{q}^*) D_p(p, \hat{q}^*) + \gamma (p - k(\hat{q}^*)) D_{pq}(p, \hat{q}^*)], \quad (4)$$

where we have already split-up the three terms that we discuss below in turn. Note that $z_{\hat{q}^*} < 0$.⁷ Hence, the sign of $d\hat{q}^*/dp$ is determined by the expression in rectangular brackets in (4). This expression comprises three different effects that the price level has on quality, which we discuss next.

⁷Precisely, this follows from inspection of the second-order derivative of firm profits, i.e., from differentiation of the first-order condition (2), and noting that $D_q > 0$ and $k' > 0$.

Relationship between Price and Quality. The term in the first term of (4) is strictly positive. When the price, and thus also the margin, is higher, the firm has more to gain by sustaining demand through choosing a higher quality. This is the most immediate effect that a higher price has on the manufacturer's incentives to maintain quality at a high level. The strength of this effect hinges on the likelihood with which quality is observed (or, likewise, on the respective fraction of informed consumers). The second term in (4) is also strictly positive from $D_p < 0$ and $k' > 0$.⁸ As the price increases and demand thereby decreases, an increase in the per-unit costs, when this is associated with higher quality, has a smaller negative impact on overall firm profits. This effect is independent of the observability of quality. We turn next to the last effect, as captured by the last term in (4). This term is zero when the marginal utility of an increase in quality is independent of the level of a consumer's valuation. This holds, for instance, when demand is linear and separable in p and q . But the effect is strictly positive when consumers with a higher absolute valuation also have a higher marginal valuation for quality.⁹ Then, as the price is higher, this increases the responsiveness of demand with respect to quality.

In what follows, we always stipulate that $D_{pq} \geq 0$, so that, altogether, from the three discussed effects, a higher price will strictly increase the firm's incentives to maintain high quality.

Lemma 1 *When $D_{pq} \geq 0$ holds, then at stage $t = 2$ in the auxiliary model quality, as obtained in (3), strictly increases with price: $d\hat{q}^*/dp > 0$.*

As we are presently only interested in the relationship between price and quality, we are silent about efficiency. When $\gamma = 1$, we can, however, directly refer to the analysis in Spence (1975) and conclude that for any given price, the resulting quality $\hat{q}^*$ is inefficiently low. As we have immediately from (3) that $\hat{q}^*$ is strictly increasing in γ , this inefficiency becomes more severe as $\gamma < 1$.

Equilibrium Price Level. To determine the price that the manufacturer chooses in $t = 1$, it is instructive to first consider the case where $\gamma = 1$, in which all consumers can observe quality. We have already noted that then $\hat{q}^*$, which is determined in (3), also

⁸This effect would be absent when, with the general cost of production $c(x, q)$, we had $c_{qx} = 0$, so that the costs of quality provision were independent of actual output.

⁹More formally, there could be a continuum of consumers, indexed by y , with respective utility qy . Then, the critical type is $\tilde{y} = p/q$, so that with distribution $F(y)$ demand is $D(p, q) = 1 - F(\tilde{y})$. When $F(\cdot)$ is the uniform distribution, we have $D_{pq} = 1/q^2 > 0$.

solves the first-order condition (2). Turning to $t = 1$, we can thus conclude that for $\gamma = 1$ the optimal price p is determined simply by setting the respective partial derivative with respect to p equal to zero. (Formally, this follows from the envelope theorem.) However, this is no longer the case when $\gamma < 1$. Then, for a given price, the resulting equilibrium quality $\hat{q}^*$ will be strictly below the value that would maximize ex-ante firm profits. The firm thus has a commitment problem vis-à-vis consumers, and the optimal choice of p takes this into account, so that now the indirect effect through the incentives to choose (higher) quality also matters.

To derive the optimal price p for general γ , in a slight abuse of notation, denote now $\pi(p, \hat{q}^*) = \pi(p, q = \hat{q}^*, \hat{q} = \hat{q}^*)$. With a slight transformation, the first-order condition in $t = 1$ becomes

$$\begin{aligned} \frac{d\pi(p, \hat{q}^*)}{dp} &= [p - k(\hat{q}^*)] D_p(p, \hat{q}^*) + D(p, \hat{q}^*) \\ &\quad + \frac{d\hat{q}^*}{dp} (1 - \gamma) [p - k(\hat{q}^*)] D_q(p, \hat{q}^*) \\ &= 0. \end{aligned} \tag{5}$$

As noted above, the second line in (5) is equal to zero when all consumers are informed, so that then (and only then) the first-order condition is obtained from the partial derivative with respect to the price.

Lemma 2 *In the equilibrium of the auxiliary model, the optimal price p that is chosen by the manufacturer in $t = 1$ solves (5).*

We again suppose that there is a unique price equilibrium. We denote this price by p^* with the corresponding quality $\hat{q}^*(p^*) = q^*$. We will make use of condition (5) in our extended model below. When we provide an illustration of the equilibrium characterization in the following section, we further utilize the presently solved auxiliary model.

3 Introducing Retailers and Negotiations

We now introduce retailers indexed by $n = 1, \dots, N$. As will become clear from the following analysis, when there is only a single retailer, the interests of the manufacturer and the single retailer (with respect to how the retail price is set) will be perfectly aligned. This is no longer the case when there are multiple retailers, even when they are not in competition with each other. In fact, below we will isolate two different sources of such

a conflict of interest: one arising from a free-riding problem and one arising from the way retail prices affect outside options and, thereby, the distribution of surplus through negotiations. For this analysis, we first have to set up and provide a solution to the bargaining problem between the manufacturer and the retailers. This is done in the present section.

Timing. Our timing, which we discuss subsequently in detail, is as follows. Retail prices, now p_n , are still set in the first period $t = 1$. Our key analysis will be the comparison of equilibrium outcomes when these are chosen individually by retailers or when they are perfectly controlled by the manufacturer (RPM). In $t = 2$ the manufacturer chooses quality. Before demand is realized, as consumers make their choice, in the presence of retailers there are now negotiations in $t = 3$. The respective solution concept is introduced below.

Our choice of timing, in which retail prices are set first, essentially deprives wholesale contracts of their steering role; these are used in $t = 3$ only to distribute profits. This implication is fully intended, and reflects two related observations about branded retail markets.

First, retail prices for branded fast-moving consumer goods are publicly committed to and visibly persistent. They are printed on packaging as suggested retail prices, advertised through weekly circulars and catalog postings, and held fixed across many shopping trips outside narrowly defined promotional windows. The macroeconomic price-setting literature documents this directly: retail price changes for typical consumer goods occur at frequencies measured in months rather than days, with substantial menu costs of revising posted prices (Nakamura and Steinsson 2008; Hosken and Reiffen 2004; see Heidhues and Köszegi 2008 for a behavioral foundation of price persistence based on loss aversion). Quality and contracting choices, by contrast, can be revisited continuously — through ingredient substitution, relaxation of hygiene or handling protocols, or renegotiation of (lump-sum) wholesale rebates that occurs even within annual framework agreements when retailers can credibly threaten to “run out of stock.”

Second, and as a consequence, posted retail prices effectively determine the stakes at the renegotiation stage, while initial wholesale contracts have limited steering power. RPM is therefore not redundant with wholesale contracting but is instead the only direct device through which a manufacturer can lock in the price level that supports the brand’s quality image.

Finally, our choice of timing also has the following additional benefit. It allows us to focus on the difference in retailers' and the manufacturer's preferences with respect to the role of the retail price as a means to communicate the quality image. In our model, this is derived fully from primitives. As we show below, with exogenously fixed quality, prices as well as the distribution of profits would not depend on the allocation of "price ownership," so that the difference is driven fully by the relationship between price and quality (image).

Motivation of a Bargaining Solution Concept. We now introduce multilateral negotiations at the beginning of $t = 3$. A large part of the literature typically assumes that one side, notably the manufacturer, can commit to making take-it-or-leave-it offers, thereby reducing all other parties to their outside options. An important shortcoming of this approach, in particular in light of our application, is that the outside option of the proposing party, and with it any (price-setting) strategies to affect it, would not come into play. De Fontenay and Gans (2014), building on Stole and Zwiebel (1996), represents one recent contribution in the literature that allows for proposals by both sides in a noncooperative (open-ended) bargaining game. There is also a literature that applies cooperative game theory to, generally speaking, bargaining in networks with externalities (that is, in our context, when retailers are in competition).

It turns out that, when applied to our setting, the general approach that we propose next yields the same outcome as when we applied the non-cooperative approach of De Fontenay and Gans (2014) or one of the existing cooperative approaches (discussed below). We feel, however, that our approach, which builds on Inderst and Wey (2003), is, at least in the present application, particularly straightforward and intuitive. To formalize it, denote the manufacturer by M and each retailer by Rn with $n = 1, \dots, N$. As we will for $t = 1$ focus on a symmetric price equilibrium and as retailers will also be otherwise symmetric, we can restrict our subsequently introduced notation to the case where at most one retailer is different (by charging a different price). Without loss of generality, we let this be retailer $R1$, so that at most $p_1 \neq p$. A possible set of agreements can thus be described by the total number of agreeing retailers ($i \leq N$) and by whether this contains $R1$. Denote by $\Pi_{In}^1(i)$ the *joint* profits of the ("Insider") coalition of i retailers that agrees with M , where the superscript 1 denotes that this contains $R1$. The respective notation when this does not contain $R1$ is $\Pi_{In}^0(i)$, i.e., with superscript 0. For all non-agreeing outsiders, denote the *individual* profit by $\pi_{Out}^1(i)$ when $R1$ agrees and when $R1$ does not agree by $\pi_{Out,R1}^0(i)$ for $R1$ and by $\pi_{Out,Rn}^0(i)$ for all other $N - i - 1$ non-agreeing retailers.

As of $t = 3$, contracts only serve the distribution of joint profits, so we can suppress a separate notation for the respective transfers and characterize the outcome directly in terms of payoffs. A bilateral agreement now conditions, as in Inderst and Wey (2003), on what we call a “contingency,” which is here the set of other retailers to which the manufacturer supplies. We can thus think of a bilateral agreement as, in our context, a vector of transfers that conditions on the manufacturer’s supply relations. While in equilibrium (cf. below) the manufacturer will supply all retailers, the transfers and resulting payoffs for all other contingencies will be payoff relevant as outside options. While we feel that such contingent contracting is also to some extent realistic, as we explore below, it can also be regarded more simply as part of a bargaining solution that thereby ensures that a party’s contribution to various subcoalitions influences its equilibrium payoff.¹⁰

For a contingency where M and $R1$ agree, denote the respective payoff of the manufacturer by $V_M^1(i)$, that of $R1$ by $V_{R1}^1(i)$, and that of the other $i - 1$ agreeing retailers by $V_{Rn}^1(i)$. When $R1$ and M do not agree, the payoff of the manufacturer is denoted by $V_M^0(i)$ and that of the agreeing retailers by $V_{Rn}^0(i)$.

Solution Concept. For each contingency, we require that the respective contracts and, thus, the resulting payoffs satisfy “balancedness”: The incremental joint payoff, that is, relative to the state without this additional agreement, must be equal for both sides. For an agreement with $R1$ this is the case if and only if

$$V_M^1(i) - V_M^0(i - 1) = V_{R1}^1(i) - \pi_{Out,R1}^0(i - 1), \quad (6)$$

which must hold for all i (i.e., for all possible other agreements, as described by the number i of all agreements). Recall also that $V_M^0(i - 1)$ represents the manufacturer’s disagreement payoff in this particular case, where there is no agreement with $R1$, while the respective disagreement payoff for $R1$ is the “outside option” $\pi_{Out,R1}^0(i - 1)$. These values are subtracted to obtain the respective incremental payoff for M (the left-hand side of (6)) and the respective incremental payoff for $R1$ (the right-hand side of (6)).

Turning to an agreement with any other retailer Rn , we must make the following distinction: For a contingency where there is also agreement with $R1$, equal division of the incremental joint payoff, now between M and Rn , requires that

$$V_M^1(i) - V_M^1(i - 1) = V_{Rn}^1(i) - \pi_{Out}^1(i - 1), \quad (7)$$

¹⁰After all, real-world bargaining typically does not follow a fixed protocol and may also vary considerably between different situations, so that, in our view, a solution concept should offer flexibility and incorporate much of the information of the specific economic situation.

while for a contingency without an agreement with $R1$, we likewise have that

$$V_M^0(i) - V_M^0(i-1) = V_{Rn}^0(i) - \pi_{Out,Rn}^0(i-1). \quad (8)$$

Conditions (6) to (8) complete the characterization of our solution concept.

The following characterization will be restricted to prices set in $t = 1$ that ensure that joint profits are maximized with the “grand coalition,” $i = N$.¹¹ In light of future calculations, we also derive a simplified characterization for the case where all retailers are symmetric, as also $p_1 = p$. Then, profits of “stand-alone” retailers, which have not agreed to be supplied by M , only depend on the number of agreeing retailers and can be more simply denoted by $\pi_{Out}(i)$, while joint profits for the agreeing coalition can, again by symmetry, be written as $\Pi_{In}^1(i) = i\pi_{In}(i)$.

Lemma 3 *When there are N retailers, suppose that at $N - 1$ retailers the same price p prevails, while at most one retailer chooses potentially a different price. Assume without loss of generality that the latter retailer is $R1$. The bargaining outcome in $t = 3$ must satisfy the imposed property of “balancedness” across all contingencies (i.e., all possible sets of agreements). Then, the manufacturer’s payoff equals*

$$V_M^* = \frac{1}{N(N+1)} \left\{ \sum_{i=0}^{N-1} (i+1) [\Pi_{In}^1(i+1) - \pi_{Out,R1}^0(i) - i\pi_{Out}^1(i)] + \sum_{i=1}^{N-1} \sum_{j=1}^i [\Pi_{In}^0(j) - j\pi_{Out,Rn}^0(j-1)] \right\} \quad (9)$$

and with symmetric retailers and thus also $p_1 = p$ abbreviates to

$$V_M^* = \frac{1}{N+1} \sum_{i=1}^N i [\pi_{In}(i) - \pi_{Out}(i-1)]. \quad (10)$$

The payoff of retailer $R1$ is

$$V_{R1}^* = V_M^* + \pi_{Out,R1}^0(N-1) - \frac{1}{N} \sum_{i=1}^{N-1} [\Pi_{In}^0(i) - i\pi_{Out,Rn}^0(i-1)], \quad (11)$$

and with symmetry it equals the payoff of all other retailers Rn and can be written as

$$V_R^* = V_M^* - \frac{1}{N} \left[\sum_{i=1}^{N-1} i\pi_{In}(i) - \sum_{i=1}^N i\pi_{Out}(i-1) \right]. \quad (12)$$

¹¹This implies, in particular, that the possibly deviating price p_1 does not fall too short of p . This restriction is without loss of generality, provided the alternative supply option is sufficiently unattractive.

Proof. See Appendix.

The composition of payoffs is particularly transparent in case of symmetry, i.e., with expressions (10) and (12). The manufacturer’s payoff from negotiations thus represents a weighted average, that is, over all possible coalitions, of the incremental value that is generated at any given retailer. With respect to $R1$, whose price choice we will later consider, in (12) we have expressed the equilibrium payoff V_{R1}^* so that the difference from that of the manufacturer, V_M^* , is immediately evident. We comment on this difference in detail below, as it will drive the difference in price setting incentives. That is, the following term will essentially describe the difference in incentives: $V_M^* - V_{R1}^*$, which is

$$\frac{1}{N} \left[\sum_{i=1}^{N-1} i\pi_{In}(i) - \sum_{i=1}^N i\pi_{Out}(i-1) \right]. \quad (13)$$

Intuitively, the manufacturer’s payoff *relative* to that of any retailer is higher when his product generates a higher incremental surplus, i.e., when it generates a higher gross surplus in each bilateral relationship (i.e., the first term in (13)) and when a retailer’s alternative is less profitable (i.e., the second term in (13)), again weighted over all possible coalitions.¹² Our subsequent analysis reveals that our qualitative insights are more general, relying on the insight that quality and quality perceptions affect retailers’ and the manufacturer’s payoffs *differently*, both with and without an agreement.

Auxiliary Case: Exogenous Quality. Suppose for a moment that quality was exogenously given. We now argue that, at $t = 1$, the same price equilibrium is obtained regardless of whether prices p_n are chosen by individual retailers or by the manufacturer. To see this, we consider a variation in p_1 . Observe now that in the difference in profits

$$V_M^* - V_{R1}^* = \frac{1}{N} \sum_{i=1}^{N-1} [\Pi_{In}^0(i) - i\pi_{Out,Rn}^0(i-1)] - \pi_{Out,R1}^0(N-1) \quad (14)$$

none of the terms depends *directly* on p_1 , as they all relate to contingencies where there is *no* agreement with retailer $R1$. Consequently, we have indeed

$$\frac{\partial}{\partial p_1} V_M^* - V_{R1}^*|_{q\text{fixed}} = 0. \quad (15)$$

¹²Navarro (2007) derives a bargaining value for general networks with externalities by extending the “fair allocation” or “balancedness” rule of Myerson (1977a/b). This requirement mirrors the equal sharing of net surplus, which we impose for all contingencies (on which the agreement conditions in our approach). Applying Navarro’s bargaining value to our specific setting yields Lemma 3. As noted above, this value is also obtained in De Fontenay and Gans (2014) from a non-cooperative approach, where an individual disagreement restarts negotiations. All these specifications thus ensure that equilibrium payoffs depend more broadly on the potential profits achieved under various possible agreements (or coalitions).

Equation (15) is the formal counterpart of the claim in our motivation: with quality exogenously fixed, retail prices feed neither actual quality nor consumer quality perceptions, and both sides therefore agree on the symmetric retail price that maximizes joint surplus. “Price ownership” is then moot. Consequently, any wedge between the manufacturer’s preferred price p_M^* and the retailer-chosen price p_R^* that we derive in Sections 4 and 5 is driven *entirely* by the price–quality–image channel, and not by other features of the bargaining environment.

4 Image Spillover

In this section, we abstract from downstream retail competition. Without competition, the profits realized at any given retailer do not depend directly on whether the manufacturer’s product is also sold at other retailers (and at what price). This allows us to isolate the aforementioned free-riding problem, on which this section focuses. We still suppose that consumers are aware of the prices that have been set at all other retailers. For instance, this could be through advertising, but also through other shopping trips that are, however, not considered to be substitutes (e.g., as the product is not storable and consumers only decide on the basis of convenience at any given instance).

As in Section 2 the first step in the analysis is to derive the equilibrium quality $\hat{q}^*$ at $t = 2$ for given prices p_n . This is accomplished by requiring that the manufacturer’s optimal choice q_{BR} equals the expectations of uninformed consumers $\hat{q}$. The only difference is that now, in the presence of retailers, q_{BR} maximizes V_M^* , as given in Lemma 3. We relegate the formal details to the proof of Proposition 1. For a given $\hat{q}^*$, as a function of prices p_n , we can next use the present specification that retailers serve separate markets to heavily simplify the expressions for payoffs and subsequently solve for equilibrium prices.

Without competition, profits realized at an individual retailer do not directly depend on choices made at other retailers; thus, we can write, even when $p_1 \neq p$, each bilateral profit as

$$\pi(p_n, \hat{q}^*) = [p - k(\hat{q}^*)] [D(p, \hat{q}^*) - k(\hat{q}^*)].$$

Note also that with competition, “outside option” payoffs (without a bilateral agreement with the manufacturer M) for each retailer are independent of prices at all other retailers and also of the manufacturer’s quality (and can thus be simply denoted by π_{Out}). With this at hand, without competition, we have from Proposition 1 for the manufacturer

$$V_M^* = \frac{1}{2} [\pi(p_1, \hat{q}^*) + (N - 1)\pi(p, \hat{q}^*) - N\pi_{Out}].$$

For $R1$ we obtain

$$V_{R1}^* = V_{R1}(N) = V_M^* + \pi_{Out} - \frac{1}{N} \sum_{i=1}^{N-1} i [\pi(p, \hat{q}^*) - \pi_{Out}],$$

so that we have for the difference in the respective price-setting incentives

$$\frac{d}{dp_1} [V_M^* - V_{R1}^*] = \frac{N-1}{2} \frac{d\hat{q}^*}{dp_1} \left[\frac{\partial}{\partial \hat{q}^*} \pi(p, \hat{q}^*) \right]. \quad (16)$$

When expression (16) is strictly positive, this implies that the manufacturer prefers a strictly higher price. Appealing to symmetry, the respective choice $p_n = p_M^*$ at $t = 1$ is then strictly higher than the equilibrium outcome when retailers individually choose prices, $p_n = p_R^*$. In the proof of Proposition 1 we show that still $\frac{d\hat{q}^*}{dp_1} > 0$, as expected from the analysis in Section 2. We also show that $\frac{\partial}{\partial \hat{q}^*} \pi(p, \hat{q}^*) > 0$ holds if and only if from $\gamma < 1$ there is a commitment problem, again in complete analogy to the analysis without retailers in Section 2. Hence, in the present case without competition between retailers, a wedge between p_M^* and p_R^* exists if and only if quality perceptions indeed matter, as quality is not perfectly observed by all consumers. Then, expression (16) captures the fact that only the manufacturer, not individual retailers, internalizes the positive effect that a change in quality and quality perceptions has on sales at all (other) retailers.

Proposition 1 *Consider the case where the manufacturer sells through N non-competing retailers. Then, the manufacturer's preferred retail price, p_M^* , is the same as that chosen individually by retailers, p_R^* , only when there is no role for quality perceptions as $\gamma = 1$. Otherwise, both price and quality are strictly higher when the manufacturer controls retail prices: $p_M^* > p_R^*$ and $q_M^* > q_R^*$.*

Proof. See Appendix.

Proposition 1 is silent about implications for welfare and consumer surplus, to which we turn below. It embodies our first channel through which a manufacturer's control over retail prices, notably through RPM, has implications for prices and quality even when retailers do not perform non-contractible actions, as the opportunism problem lies between the manufacturer and consumers, and even when intrabrand competition is not an issue, as seen by the fact that with fixed quality, prices do not differ ($p_M^* = p_R^*$). The reason is a free-riding problem of retailers regarding the effect that individual retail prices have on quality perceptions. This is the essence of the “image theory” of RPM, as discussed in the Introduction, where in our model “image” relates to quality perceptions.

Note next that the manufacturer's preferred choice, p_M^* , is not affected by the number of non-competing retailers, while the retailers' free-riding problem is aggravated as their number N increases.

Corollary 1 *When quality perceptions matter as $\gamma \in (0, 1)$, the difference $p_M^* - p_R^* > 0$ is strictly increasing in N . Consequently, also the difference in equilibrium quality, $q_M^* - q_R^* > 0$, is strictly increasing in N .*

Proof. See Appendix.

It is worthwhile to note that while p_M^* maximizes the manufacturer's profits, the price that prevails under retailer "price ownership," p_R^* , does not maximize the joint profits of retailers. In fact, retailers would jointly be best off when setting $p_n = p_M^*$, i.e., exactly the price that would prevail under manufacturer "price ownership"! As is the essence of a free-riding problem, they, however, have a private incentive to deviate by setting a strictly lower price in their own market. The coincidence of p_M^* and the symmetric price that would maximize total retailer profits no longer holds when we introduce retailer competition in Section 5.

Welfare Comparison. We now consider the case of linear demand $D(p_n, q) = q - p_n$ and constant marginal cost of production $\frac{1}{2k}q^2$. We relegate a full derivation of the expressions to the subsequent proof. For $t = 2$, we now obtain explicitly

$$\hat{q}^* = \frac{\bar{p} + \sqrt{\bar{p}^2 + \bar{p}2\gamma(2 + \gamma)k}}{2 + \gamma}. \quad (17)$$

where $\bar{p} = \frac{1}{N} \sum_{n=1}^N p_n$. The first-order condition for $p_n = p = p_R^*$ becomes

$$\left(\frac{1}{N}\right) \frac{(\gamma k + \hat{q}^*) \left(p - \frac{\hat{q}^*(3\hat{q}^* - 2p)}{2k}\right)}{(\gamma + 2)\hat{q}^* - p} + \frac{(\hat{q}^*)^2}{2k} - 2p + \hat{q}^* = 0. \quad (18)$$

The respective condition for $p_n = p = p_M^*$ is also obtained from (18) when setting $N = 1$. This follows simply because, first, with $N = 1$ there is no difference between the two cases, and second, p_M^* does not depend on N , as noted above.

For a particular numerical specification, Figure 1 plots the two prices p_M^* and p_R^* as a function of the degree of transparency about quality, γ . For this example only, we have maximized the difference between the two scenarios by letting $N \rightarrow \infty$. Then, the

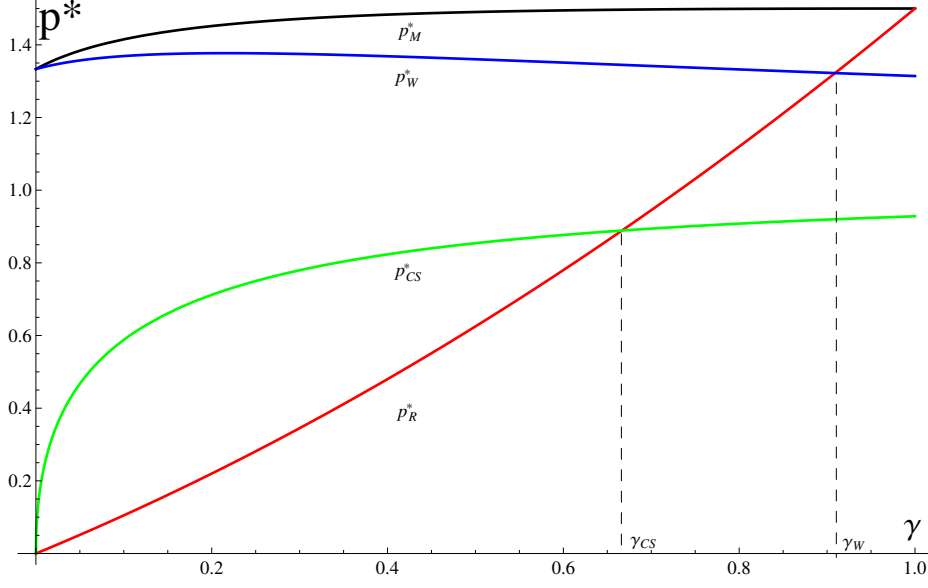


Figure 1: This figure shows p_M^* and p_R^* , next to the prices that maximize welfare, p_W^* , and consumer surplus, p_{CS}^* , as a function of γ . Parameter values are $k = 2$ and $N = \infty$.

retailers' equilibrium choice of p_R^* does not internalize all implications that prices have on quality perception at all.

Prices are always the same when $\gamma = 1$, as then quality perceptions do not matter. For all lower values of $\gamma > 0$ we have $p_M^* > p_R^*$. In Figure 1, we have also characterized the prices that would maximize total welfare and consumer surplus, p_W^* and p_{CS}^* . Intuitively, it holds that $p_W^* > p_{CS}^*$, as total welfare is not affected by mere transfers from consumers to firms. Further, at $\gamma = 1$ note that both efficiency benchmark prices are strictly lower than the price that would be obtained in the market, $p_M^* = p_R^*$. We now compare the retailers' individually preferred price p_R^* with the two efficiency benchmarks. As γ becomes sufficiently low, p_R^* falls below both the price p_W^* that would maximize total welfare (when $\gamma < \gamma_W$) and the lower price p_{CS}^* that would maximize consumer surplus (when $\gamma < \gamma_{CS}$). The deadweight loss that a higher price would entail, and even the direct loss to consumers, at least for price increases that are not too large, is more than compensated by the resulting increase in equilibrium quality. Proposition 2 shows that these results hold generally for our specification.

The implications for consumer surplus are also brought out more clearly in Figure 2. The non-monotonic line plots the difference between consumer surplus at p_R^* and p_{CS}^* , $CS(p_R^*) - CS(p_{CS}^*)$. It is zero at $\gamma = 0$, where demand becomes zero, and at $\gamma = \gamma_{CS}$, while it is otherwise strictly negative. More important is the intersection with the difference

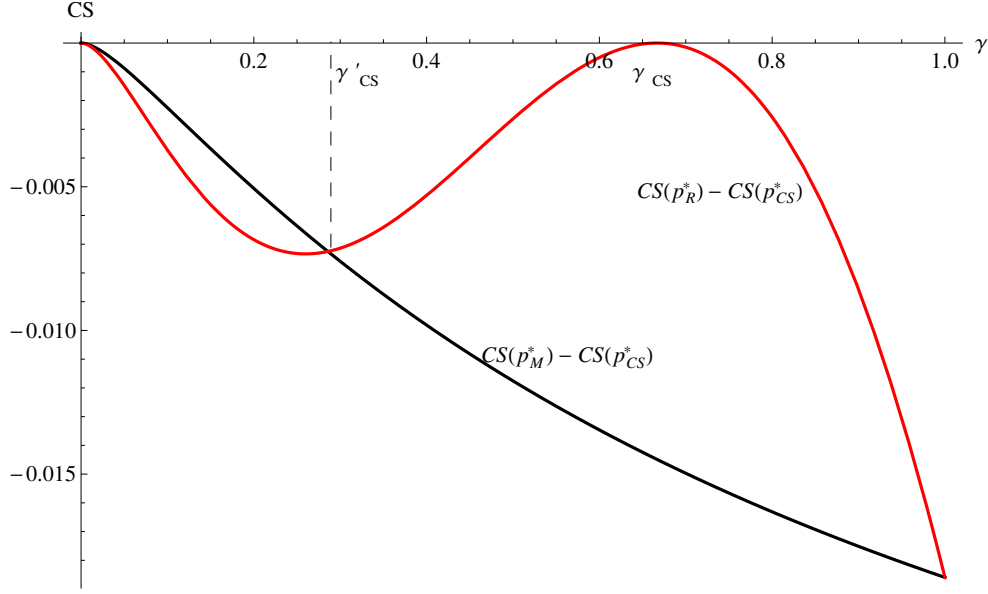


Figure 2: Consumer surplus for p_M^* and p_R^* , net of maximum consumer surplus at p_{CS}^* , as a function of γ . Parameter values are $k = 2$ and $N = \infty$.

$CS(p_M^*) - CS(p_{CS}^*)$, indicating that $CS(p_M^*) > CS(p_R^*)$ holds for low levels of transparency $0 < \gamma < \gamma'_{CS}$ and $CS(p_M^*) < CS(p_R^*)$ holds for high levels of transparency $\gamma'_{CS} < \gamma < 1$. While to save space we do not plot this separately, the same picture is obtained when we consider total welfare: For all γ below some threshold γ'_W , welfare is higher when the manufacturer controls the price. This threshold is intuitively (considerably) higher than γ'_{CS} .

Proposition 2 *In the chosen linear-quadratic specification, the following comparison of prices and efficiency (in any given market) holds:*

- i) For low levels of transparency γ , the price that prevails when retailers set prices is strictly below both the price that would maximize consumer surplus and the price that would maximize total welfare ($p_R^* < p_{CS}^*$ and $p_R^* < p_W^*$), while for high values it is strictly higher than both ($p_R^* > p_{CS}^*$ and $p_R^* > p_W^*$). The price prevailing when the manufacturer sets prices is always strictly higher than both efficiency benchmarks ($p_M^* > p_{CS}^*$ and $p_M^* > p_W^*$).*
- ii) For low levels of transparency γ , welfare and consumer surplus are strictly higher under the price that the manufacturer sets ($W(p_M^*) > W(p_R^*)$ and $CS(p_M^*) > CS(p_R^*)$), while for high values, the opposite holds such that welfare and consumer surplus are strictly higher under the price that prevails when retailers set prices ($W(p_M^*) < W(p_R^*)$ and $CS(p_M^*) < CS(p_R^*)$).*

Proof. See Appendix.

Together, the comparison of consumer surplus and welfare reveals two key insights. The first insight is that from the perspectives of both consumer surplus and welfare, efficiency can be higher under p_M^* , even though this exceeds p_R^* . The reason is the thereby induced increase in quality. The second insight is the role of quality perceptions and transparency. In the linear-quadratic specification, the free-riding problem under retailer “price ownership” becomes sufficiently important when the fraction of consumers who do not observe quality is sufficiently large. In this case, consumer surplus and total welfare are both strictly lower when retailers individually choose prices in their own interest. The opposite holds when transparency is sufficiently high, so that perceptions matter less.

5 Retailer Competition

Given our preceding analysis, we turn right to the choice of prices in $t = 1$. Recall that we are interested in the difference in price setting incentives between the manufacturer and individual retailers. As we already observed that for some exogenously fixed quality level incentives are the same, again only the indirect effect matters, i.e., the effect that prices have on quality. That is:

$$\begin{aligned} \frac{d}{dp_1} [V_M^* - V_{R1}^*] &= \frac{d\hat{q}^*}{dp_1} \frac{d}{d\hat{q}^*} [V_M^* - V_{R1}^*] \\ &= \frac{d\hat{q}^*}{dp_1} \left[\frac{1}{N} \sum_{i=1}^{N-1} i \frac{d}{d\hat{q}^*} \pi_{In}(i) - \frac{1}{N} \sum_{i=1}^N i \frac{d}{d\hat{q}^*} \pi_{Out}(i-1) \right]. \end{aligned} \quad (19)$$

We assume again concavity of profits, now of $\pi_{In}(i)$ and $\pi_{Out}(i)$ in $\widehat{q}^*$. The second sum in (19) captures the effects that a higher induced quality level has on the profits of non-agreeing retailers under various contingencies (i.e., depending on the number of agreeing retailers).¹³ Without competition, these terms were equal to zero. With competition, these terms generate a wedge between the preference of the manufacturer and that of any retailer when

$$\frac{d}{d\widehat{q}^*}\pi_{Out}(i) < 0 \text{ for } 1 \leq i \leq N - 1,$$

as then a higher quality of the manufacturer's product undermines the "outside option" of a non-agreeing retailer. This makes the manufacturer prefer a higher retail price, as the thereby induced higher quality reduces retailers' profits when they do not come to an agreement with the manufacturer. This ultimately benefits the manufacturer. The presently discussed effect thus isolates another mechanism through which a conflict of interest between retailers and the manufacturer can lead to different preferences regarding the retail price.

Without competition, we already know that the first term in (19) is zero when $\gamma = 1$ and otherwise strictly positive ("free riding" problem). More generally, it captures the effect that quality has on the joint profits of agreeing retailers, again summed up over all possible contingencies; that is, whether one or up to $N - 1$ retailers agree.¹⁴ At first, it may seem intuitive that all expressions $\frac{d}{d\widehat{q}^*}\pi_{In}(i)$ should be positive, i.e., that higher quality increases *insiders'* profits, in particular when there is competition. However, this ignores the fact that the provision of quality is costly. To sign also this term and thus ultimately the whole expression, we undertake two separate analyses. We first derive general clear-cut implications when the level of transparency is low, so that perceptions matter a lot. Then, we derive implications for the specific case with linear demand.

Before turning to these analyses, the overall nature of the game considered in this paper should be recalled, as the present "reduced form" representation may mask the underlying complexity. We consider and compare the equilibrium outcomes when either the manufacturer or all retailers simultaneously set retail prices in $t = 1$, which affect demand both directly and indirectly through the induced quality equilibrium. The latter, in turn, is shaped by both the manufacturer's incentives, which depend on the outcome of the subsequent negotiations with all N retailers, as well as the quality perceptions that

¹³Note for completeness that clearly $\frac{d}{d\widehat{q}^*}\pi_{Out}(0) = 0$ as then there is no retailer that offers the incumbent manufacturer's product.

¹⁴The respective term for N agreeing retailers dropped out as we consider the difference in profits between the manufacturer and a retailer.

the choice of all prices together induce at consumers who do not directly observe quality. Perceptions and the actual chosen quality must, of course, coincide in equilibrium.

Transparency. Recall that for $\gamma < 1$ there is an opportunism problem vis-à-vis consumers who do not directly observe quality. Precisely, while for given prices the choice of q_{BR} maximizes the manufacturer's profits, V_M^* , this is not the case for the resulting equilibrium quality $\hat{q}^*$. While in equilibrium it holds that $\hat{q}^* = q_{BR}$, $\hat{q}^*$ only satisfies the respective first-order condition when $\gamma = 1$. Consider now instead the auxiliary problem to choose quality $\hat{q}^*$ so as to maximize the profits of an agreeing coalition with i retailers, $i\pi_{In}(i)$. As γ becomes sufficiently small, the equilibrium quality is always too low, given the opportunism problem, regardless of the choice of $i = 1, \dots, N$. Appealing to concavity, the first term in (19) is then strictly positive.

Proposition 3 *Consider the general case where retailers can be in competition. When $\gamma > 0$ is sufficiently small, such that a large fraction of consumers must rely on quality perceptions, the manufacturer's preferred retail price and the thereby induced quality are both strictly higher than when retailers individually choose their preferred retail price.*

Hence, when transparency is low, such that quality *perceptions* matter a lot, we can sign both terms in expression (19). The manufacturer's optimal choice of the retail price will then be strictly higher than that of retailers, implying also a strictly higher quality, *both* because the manufacturer cares about the positive impact of quality on his bargaining position (the second sum in (19)) *and* because the manufacturer cares more about the problem of underprovision of quality due to the opportunism problem vis-à-vis consumers (the first term in (19)).

Linear Demand. We now return to our specification with linear demand, for which we extend the quadratic utility function to the case with N differentiated retailers. Demand

at retailer n is given by¹⁵

$$(q_n - p_n) - \delta \sum_{m \neq n} (q_m - p_m). \quad (20)$$

We are now specific about the choices made by non-agreeing retailers. Also applying the same timing of moves, we suppose that then consumers can buy a good of fixed quality q_0 and price p_0 at these retailers. Considering again symmetric prices for the manufacturer's products, joint profits generated with one of i agreeing retailers are then given by

$$\pi_{In}(i) = [(1 - \delta(i - 1))(\hat{q}^* - p) - \delta(N - i)(q_0 - p_0)](p - k(\hat{q}^*)).$$

Profits for each of the $(N - i)$ non-agreeing retailers are given by

$$\pi_{Out}(i) = [(1 - \delta(N - i - 1))(q_0 - p_0) - \delta i(\hat{q}^* - p)](p_0 - k_0).$$

Proposition 4 *Consider the case with linear demand and competition. Then for all $\gamma > 0$ the manufacturer's preferred retail price and the thereby induced quality are both strictly higher than when retailers individually choose their preferred retail price.*

Proof. See Appendix.

Incidentally, competition can now also lead to “gold-plating” in terms of quality by the manufacturer, as he thereby improves his bargaining position. That is, the quality choice induced by the higher price set by the manufacturer strictly exceeds an efficiency benchmark. To illustrate this, take $\gamma = 1$, such that equilibrium quality maximizes V_M^* , and, by the envelope theorem, that the manufacturer's preferred price is determined by setting the partial derivative with respect to price equal to zero. With the linear demand specification, note that a marginal price reduction has the same demand expanding effect as a marginal increase in quality. Therefore, by optimality, also the marginal cost of an increase in quality ($k'(q)$) must also equal the “marginal cost” of a price reduction, which is equal to one. Using again $k(q) = \frac{1}{2k}q^2$, this yields $q_M^* = k$ (provided that $p = p_M^*$). Note next that total welfare is now

$$W = N(1 - \delta) \left[\frac{1}{2}(q - p)^2 + (q - p)(p - k(q)) \right].$$

¹⁵That is, a representative consumer derives utility of $u(\mathbf{x}, q) = \sum_{n=1}^N [q_n x_n - \frac{1}{2}\beta x_n^2] - \phi \sum_{m \neq n} x_n x_m$, where x_n denotes the quantity bought at retailer n . Setting $\partial u / \partial x_n = p_n$ and solving for x_n yields (20), where $\delta = \phi / [(\beta - \phi)(\beta + (N - 1)\phi)]$ and β and ϕ are such that $\frac{\beta + (N - 2)\phi}{(\beta - \phi)(\beta + (N - 1)\phi)} = 1$. The quadratic-in-quantity representative-consumer formulation should be understood as a reduced-form aggregation of a population of heterogeneous consumers each choosing a single retailer in a differentiated discrete-choice problem (cf., e.g., Anderson, de Palma, and Thisse 1992); the within-retailer concavity reflects heterogeneity across consumers in their preferred outlet rather than literal diminishing marginal utility from buying more units of the same product at the same retailer.

For a given symmetric price p , the welfare maximizing quality would thus be

$$q_W = \frac{2}{3}(k + p),$$

which is strictly decreasing in p . Evaluated at $p = p_M^*$, as this is strictly decreasing in δ ,¹⁶ this suggests that quality may be indeed too high when there is sufficient competition. This can be confirmed with the respective threshold

$$\delta'' = \left(\frac{3}{2(N-1)} \right) \frac{k}{k + q_0 - p_0 + 2(p_0 - k_0)}.$$

Still, despite such “gold plating,” here for $\delta > \delta''$, we can show that the outcome may still be more efficient than when price and quality are lower under the respective price choices of retailers.¹⁷ However, in the linear specification, there is no monotonic relationship between the welfare comparison and the degree of competition. Still, our observations in this section further confirm the paper’s main message of identifying a channel-coordination rationale for RPM through the described “image theory,” though we cannot provide guidance as to whether this should be more or less applicable depending on the degree of downstream competition.¹⁸

6 Concluding Remarks

We consider a manufacturer’s incentives to choose quality in an environment where this is not directly observed by all consumers, so that their quality perceptions matter. We further ask how these incentives are influenced by the product’s price. Here, we derive various channels through which a higher price can induce higher quality and quality perception by those consumers who do not directly observe quality but form rational expectations. This set-up is then embedded into a game where either retailers or the manufacturer control the retail price (“price ownership”).

We isolate different sources of a conflict of interest between retailers and the manufacturer, which lead to different prices and qualities depending on which side exerts control

¹⁶Precisely: $\frac{dp_M^*}{d\delta} = \frac{-3}{2(3-2\delta)^2} [(q_0 - p_0) + 2(p_0 - k_0)]$.

¹⁷Such an example is obtained for $k = 2$, $N = 2$, as well as $p_0 = 0.5$, $q_0 = 1.5$, and $k_0 = 0.1$ for the outside option, in which case we have $\delta'' \approx 0.79$. Welfare is then higher under p_M^* than under p_R^* if $\delta > 0.24$.

¹⁸It may be conjectured that in the presence of upstream competition, there may be less scope for welfare losses resulting from higher prices. However, the interaction of asymmetric information about quality, up- and downstream competition, and multilateral negotiations has so far proved to be insufficiently tractable. Modifying our approach or complementing the analysis with a different modeling set-up should be an interesting avenue for future research.

over prices. One channel that supports such a conflict of interest works through a reputation spill-over across retailers. Each individual retailer does not take into account how his price choice affects the overall perception of the product’s quality and, thereby, also the equilibrium quality choice. Then, “price ownership” of the manufacturer leads to higher quality. Though this is associated with a higher price, we identify instances when both efficiency and consumer surplus are still higher. A second channel supporting a conflict of interest between the manufacturer and retailers operates when there is competition between retailers. Then, a higher (perceived) quality, which would be induced by a higher retail price, reduces a retailer’s but increases the manufacturer’s outside option in bilateral negotiations. Again, the manufacturer’s preferred price and the thereby induced equilibrium quality are higher than when retailers control prices.

Our model rests on a particular timing of strategies, in which retail prices are committed to before quality and contracting decisions. We motivated this by the observed persistence of posted retail prices for branded fast-moving consumer goods — visible on packaging, advertised circulars, and catalog postings, with menu costs that anchor prices over months at a time — relative to the comparatively continuous adjustment of quality and renegotiation of wholesale terms. The key implication that derives from this specification is that retail prices, once committed, shape the manufacturer’s subsequent quality incentives and consumers’ resulting quality perceptions; RPM is then the direct device through which the manufacturer locks in the price level that supports the brand’s quality image. Future work could model the dynamic interplay between wholesale and retail prices and the continuous choice of quality, taking into account that all these choices are to some extent (and only to some extent) persistent. In particular, quality and quality perceptions would then be the result of current as well as past price choices at all retailers.

7 Appendix: Omitted Derivations and Proofs

Proof of Lemma 3. For contingencies without an agreement with $R1$, we have the joint profit condition

$$V_M^0(i) + iV_{Rn}^0(i) = \Pi_{In}^0(i),$$

which, together with (8), yields

$$(i + 1)V_M^0(i) = iV_M^0(i - 1) + [\Pi_{In}^0(i) - i\pi_{Out,Rn}^0(i - 1)].$$

From this, we obtain the following for the manufacturer:

$$V_M^0(i) = \frac{1}{i+1} \sum_{j=1}^i [\Pi_{In}^0(j) - j\pi_{Out,Rn}^0(j-1)]. \quad (21)$$

When there is an agreement with R_1 , we have the joint profit condition

$$V_M^1(i) + (i-1)V_{Rn}^1(i) + V_{R1}^1(i) = \Pi_{In}^1(i).$$

Here, we can substitute (6) and (7), which yields

$$V_M^1(i) = \frac{1}{i(i+1)} \sum_{j=0}^{i-1} (j+1) [\Pi_{In}^1(j+1) - j\pi_{Out}^1(j) - \pi_{Out,R1}^0(j) + V_M^0(j)].$$

Making use of (21) we obtain $V_M^1(i)$ and, together with (6) also $V_{R1}^1(i)$. Choosing $i = N$ yields the final characterization.¹⁹ **Q.E.D.**

Proof of Proposition 1. Denote by $\pi(p_n, q, \hat{q})$ the bilateral profits for an agreement with retailer n , which depend only on the respective price p_n as well as on actual and perceived quality. We will first make use of this notation to derive equilibrium quality at stage $t = 2$, $\hat{q}^* = \hat{q} = q$. Subsequently, as done in the main text, we can further simplify notation by substituting for this choice. At $t = 2$, the manufacturer chooses q to maximize his own payoff. With the preceding observations and making use of Lemma 3, without competition we have

$$V_M^* = \frac{1}{2} \left[\sum_{n=1}^N \pi(p_n, q, \hat{q}) - N\pi_{Out} \right], \quad (22)$$

so that

$$q_{BR} = \arg \max_q \sum_{n=1}^N \pi(p_n, q, \hat{q}).$$

Or, with the first-order condition as in (2), q_{BR} solves

$$\gamma \left[\sum_{n=1}^N (p_n - k(q_{BR})) D_q(p_n, q_{BR}) \right] - k'(q_{BR}) \sum_{n=1}^N [\gamma D(p_n, q_{BR}) + (1-\gamma) D(p_n, \hat{q})] = 0. \quad (23)$$

Recall that at this stage we solve for the equilibrium where a consumer who does not observe quality holds rational beliefs, so that $q_{BR} = \hat{q} = \hat{q}^*$. Note again that profits in one retail market currently depend on the price in another retail market only through the

¹⁹Again, we suppose here that the incremental profit for each agreement is, for all contingencies, positive.

effect that the other price has on the manufacturer's incentives to adjust quality—and, for $\gamma < 1$, on the respective beliefs of uninformed consumers. Hence, the equilibrium requirement for $\hat{q}^*$ at $t = 2$ is that, in analogy to condition (3),

$$z(p_n, \hat{q}^*) := \gamma \left[\sum_{n=1}^N (p_n - k(\hat{q}^*)) D_q(p_n, \hat{q}^*) \right] - k'(\hat{q}^*) \sum_{n=1}^N D(p_n, \hat{q}^*) = 0, \quad (24)$$

so that from implicit differentiation

$$\frac{d\hat{q}^*}{dp_n} = \frac{1}{-z_{\hat{q}^*}} [\gamma D_q(p_n, \hat{q}^*) - k'(\hat{q}^*) D_p(p_n, \hat{q}^*) + \gamma (p_n - k(\hat{q}^*)) D_{pq}(p_n, \hat{q}^*)].$$

As in Lemma 1 this is strictly positive when, where we have rearranged terms,

$$\begin{aligned} z_{\hat{q}^*} = & \gamma \left[\sum_{n=1}^N [(p_n - k(\hat{q}^*)) D_{qq}(p_n, \hat{q}^*) - 2k'(\hat{q}^*) D_q(p_n, \hat{q}^*)] \right] \\ & - k''(\hat{q}^*) \sum_{n=1}^N D(p_n, \hat{q}^*) - (1 + \gamma) k'(\hat{q}^*) \sum_{n=1}^N D_q(p_n, \hat{q}^*) < 0. \end{aligned}$$

This holds because the first term is strictly negative due to the concavity of the program for q_{BR} , and the second term is strictly negative from $k' > 0$ and $D_q > 0$.

For the manufacturer's optimal price choice, we obtain

$$\begin{aligned} \frac{dV_M^*}{dp_n} = & D_p(p_n, \hat{q}^*) (p_n - k(\hat{q}^*)) + D(p_n, \hat{q}^*) \\ & + \sum_{n'=1}^N [D_q(p_{n'}, \hat{q}^*) (p_{n'} - k(\hat{q}^*)) - D(p_{n'}, \hat{q}^*) k'(\hat{q}^*)] \frac{d\hat{q}^*}{dp_n} = 0. \end{aligned} \quad (25)$$

When some retailer n (notably $R1$) could optimally choose p_n , after substituting into Lemma 3, this would satisfy the requirement

$$\begin{aligned} \frac{dV_{Rn}^*}{dp_n} = & D_p(p_n, \hat{q}^*) (p_n - k(\hat{q}^*)) + D(p_n, \hat{q}^*) \\ & + [D_q(p_n, \hat{q}^*) (p_n - k(\hat{q}^*)) - D(p_n, \hat{q}^*) k'(\hat{q}^*)] \frac{d\hat{q}^*}{dp_n} = 0. \end{aligned} \quad (26)$$

For $\gamma = 1$, we have from (23) that

$$D_q(p_n, \hat{q}^*) (p_n - k(\hat{q}^*)) - D(p_n, \hat{q}^*) k'(\hat{q}^*) = 0,$$

so that from the respective first-order conditions and focusing on a symmetric outcome, we have $p_M^* = p_R^*$. For $\gamma < 1$, note first that, again at the symmetric choice $p_n = p$, it holds that

$$D_q(p_n, \hat{q}^*) (p_n - k(\hat{q}^*)) - D(p_n, \hat{q}^*) k'(\hat{q}^*) > 0,$$

from which, together with (26) and (25) as well as $\frac{d\hat{q}^*}{dp_n} > 0$, we have in this case that $p_M^* > p_R^*$. **Q.E.D.**

Proof of Corollary 1. An immediate and intuitive way to see that p_M^* is independent of N is to maximize V_M^* directly with respect to a symmetric price p . Note first that at symmetric prices $p_n = p$, we have that $\frac{d\hat{q}^*}{dp} = N \frac{d\hat{q}^*}{dp_n}$ and further that $\frac{d\hat{q}^*}{dp}$ is simply obtained from, with a slight abuse of notation,

$$z(p, \hat{q}^*) = \gamma [(p - k(\hat{q}^*)) D_q(p, \hat{q}^*)] - k'(\hat{q}^*) D(p, \hat{q}^*) = 0.$$

With this at hand, the first-order condition for $p = p_M^*$ can be written as

$$D_p(p, \hat{q}^*) (p - k(\hat{q}^*)) + D(p, \hat{q}^*) + [D_q(p, \hat{q}^*) (p - k(\hat{q}^*)) - D(p, \hat{q}^*) k'(\hat{q}^*)] \frac{d\hat{q}^*}{dp} = 0 \quad (27)$$

and is thus indeed independent of N . We next argue that, instead, p_R^* is strictly decreasing in N . Using the concavity of the program, from (26), this is the case when, at symmetric prices $p_n = p = p_R^*$, this holds for $\frac{d\hat{q}^*}{dp_n}$. As z_{p_n} is independent of N , we only need to verify that $|z_{\hat{q}^*}|$ is increasing, which holds as, using $p_n = p$, it is proportional to N . **Q.E.D.**

Proof of Proposition 2. The joint profits of the manufacturer along with retailer n are given by

$$\pi(p_n, q, \hat{q}) = (q_\gamma - p_n) \left(p_n - \frac{1}{2k} q^2 \right),$$

where we use $q_\gamma = \gamma q + (1 - \gamma) \hat{q}$. The linear demand structure allows us to write the equilibrium condition for $\hat{q}^*$ in (24) in terms of the average retail price $\bar{p} = \frac{1}{N} \sum_{n=1}^N p_n$:

$$\gamma \left(\bar{p} - \frac{1}{2k} (\hat{q}^*)^2 \right) - \frac{1}{k} \hat{q}^* (\hat{q}^* - \bar{p}) = 0, \quad (28)$$

which can be solved for $\hat{q}^*$ as given by (17) and used in the first-order condition (18). Further, to calculate the example, for $N \rightarrow \infty$ expression (18) simplifies to

$$\frac{(\hat{q}^*)^2}{2k} - 2p_R^* + \hat{q}^* = 0.$$

To compare efficiency, we obtain the consumer surplus *in each market* from the utility function of the respective representative consumer.

$$CS = \frac{1}{2} (q - p)^2$$

and total welfare

$$W = \frac{1}{2} (\hat{q}^* - p)^2 + (\hat{q}^* - p) (p - k(\hat{q}^*)).$$

Consider assertion i). We will first show that $p_M^* > p_{CS}^*$. To this end, rewrite condition (27) for $D(p, \hat{q}^*) = \hat{q}^* - p$ to obtain

$$\left. \frac{d\hat{q}^*}{dp} \right|_{p_M^*} = \frac{(p - k(\hat{q}^*)) - (\hat{q}^* - p)}{(p - k(\hat{q}^*)) - (\hat{q}^* - p) k'(\hat{q}^*)} < 1. \quad (29)$$

This follows from (17) which implies that $D(p, \hat{q}^*) > 0$ only if $p < 2k$, such that $\hat{q}^* < 2k$ and thus $k'(\hat{q}^*) < 1$. The assertion that $p_M^* > p_{CS}^*$ then follows from

$$\left. \frac{dCS}{dp} \right|_{p_M^*} = (\hat{q}^* - p) \left(\left. \frac{d\hat{q}^*}{dp} \right|_{p_M^*} - 1 \right) < 0. \quad (30)$$

Recall next from the proof of Proposition 1 that $p_R^* = p_M^*$ for $\gamma = 1$, which implies that also $p_R^* > p_{CS}^*$ for large values of γ . Consider next the opposite end: $\gamma = 0$. There, we have $p_{CS}^* = 0$ and $\frac{dp_{CS}^*}{d\gamma} \rightarrow \infty$ as $\gamma \rightarrow 0$.²⁰ Likewise, for $\gamma = 0$ we obtain $p_R^* = 0$ and

$$\left. \frac{dp_R^*}{d\gamma} \right|_{\gamma=0} = \frac{k \left(3\sqrt{N(9N-8)} - \sqrt{\frac{9N-8}{N}} + 9 - 7N \right)}{4(N-1)},$$

which is equal to $5/4(\sqrt{5} - 1)k$ for $N = 2$, it strictly decreases in N , and it converges to $k/2$ as $N \rightarrow \infty$. Given the boundedness of $\frac{dp_R^*}{d\gamma}$, we have thus shown that $p_{CS}^* > p_R^*$ must hold for small values of γ , regardless of the choice of N . (In fact, for $N \rightarrow \infty$, one can even solve in closed form for the threshold $\gamma_{CS} = \frac{2}{3}$.)

Furthermore, from (17), we obtain for $\gamma = 0$ that $\hat{q}^* = p$, and thus $\frac{d\hat{q}^*}{dp} = 1$, which in turn implies from (30) that $\left. \frac{dCS}{dp} \right|_{p_M^*} = 0$; so that since $W = CS + \pi$, we get $p_W^* = p_M^*$. Furthermore, solving the system of (18) (with $N = 1$) and (28) for the equilibrium values p_M^* and q_M^* yields that $p_M^* = \frac{2}{3}k$ for $\gamma = 0$. Hence, we have that $p_W^* > p_R^*$ for small values of γ . Since $W = \pi + CS$ it follows from (30) that $p_W^* < p_M^*$ for all $\gamma > 0$ and therefore, since $p_R^* = p_M^*$ for $\gamma = 1$, also $p_W^* < p_R^*$ for large values of γ .

Now turn to assertion ii) and consider again $\gamma = 1$ where it holds that $p_R^* = p_M^*$, so that $CS(p_R^*) = CS(p_M^*)$ (and also $W(p_R^*) = W(p_M^*)$). However, since at $\gamma = 1$, it holds that

$$\frac{dW(p_R^*)}{d\gamma} = \frac{dCS(p_R^*)}{d\gamma} = \left(\frac{1}{N} \right) \frac{k^2}{36} < \frac{k^2}{36} = \frac{dCS(p_M^*)}{d\gamma} = \frac{dW(p_M^*)}{d\gamma},$$

it follows that $CS(p_M^*) < CS(p_R^*)$ and $W(p_M^*) < W(p_R^*)$ for high values of γ .

²⁰Precisely: $p_{CS}^* = (1 + \gamma) \sqrt{\gamma(2 + \gamma)}k - \gamma(2 + \gamma)k$.

Now consider $\gamma = 0$, where $CS(p_R^*) = CS(p_M^*) = 0$, and further, $\frac{dCS(p_R^*)}{d\gamma} = \frac{dCS(p_M^*)}{d\gamma} = 0$. However, $\frac{d^2CS(p_M^*)}{d\gamma^2} = \frac{4}{9}k^2$ and

$$\frac{d^2CS(p_R^*)}{d\gamma^2} = k^2 \frac{(\sqrt{9N-8} - 5\sqrt{N})^2}{16N},$$

which, for $N = 2$ equals $0.477k^2 > \frac{4}{9}k^2$, but for $N = 3$ this equals $0.385k^2 < \frac{4}{9}k^2$. Furthermore, $\frac{d^2CS(p_R^*)}{d\gamma^2}$ is strictly decreasing in N and converges to $1/4k^2$ as $N \rightarrow \infty$. Therefore, if $N > 2$, $CS(p_R^*) < CS(p_M^*)$ for small values of γ . Finally, we have $W(p_R^*) = W(p_M^*) = 0$, but $\frac{dW(p_R^*)}{d\gamma} = 0$ and $\frac{dW(p_M^*)}{d\gamma} = \frac{8}{27}k^2$ at $\gamma = 0$, so that $W(p_R^*) < W(p_M^*)$ for small values of γ . **Q.E.D.**

Proof of Proposition 4. With linear demand, we have

$$V_M^* = \frac{N}{6} \{ [(3 - 2\delta(N-1))(\hat{q}^* - p) - \delta(N-1)(q_0 - p_0)] [p - k(\hat{q}^*)] \\ - [(3 - \delta(N-1))(q_0 - p_0) - 2\delta(N-1)(\hat{q}^* - p)] [p_0 - k_0] \}$$

and

$$V_M^* - V_{R1}^* = \frac{(N-1)}{6} [(3 - 2\delta(N-2))(\hat{q}^* - p) - \delta(N-1)(q_0 - p_0)] [p - k(\hat{q}^*)] \\ - \frac{(N+1)}{6} [(3 - \delta(N-1))(q_0 - p_0) - 2\delta(N-1)(\hat{q}^* - p)] [p_0 - k_0]. \quad (31)$$

Now observe that from the first-order condition for q_{BR} we obtain, with $q_{BR} = \hat{q} = \hat{q}^*$, that

$$(3 - 2\delta(N-1)) [\gamma(p - k(\hat{q}^*)) - (\hat{q}^* - p)k'(\hat{q}^*)] \\ = -\delta(N-1) [(q_0 - p_0)k'(\hat{q}^*) + 2\gamma(p_0 - k_0)].$$

This implies that

$$(3 - 2\delta(N-1)) [(p - k(\hat{q}^*)) - (\hat{q}^* - p)k'(\hat{q}^*)] \\ \geq -\delta(N-1) [(q_0 - p_0)k'(\hat{q}^*) + 2(p_0 - k_0)], \quad (32)$$

with a strict inequality for $\gamma < 1$. Differentiating (31) with respect to q yields

$$\left. \frac{d(V_M^* - V_{R1}^*)}{dq} \right|_{\hat{q}^*} = \frac{(N+1)}{6} \left\{ \left(\frac{N-1}{N+1} \right) (3 - 2\delta(N-2)) [p - k(\hat{q}^*) - (\hat{q}^* - p)k'(\hat{q}^*)] \right. \\ \left. + \delta(N-1) (q_0 - p_0)k'(\hat{q}^*) + 2\delta(N-1) [p_0 - k_0] \right\},$$

from which, by using (32) and after some transformations, we obtain

$$\left. \frac{d(V_M^* - V_{R1}^*)}{dq} \right|_{\hat{q}^*} \geq \delta(N-1) \left(\frac{6 - 5\delta(N-1)}{3 - 2\delta(N-1)} \right) [(q_0 - p_0) k'(\hat{q}^*) + 2(p_0 - k_0)] > 0,$$

where by (32) the first inequality holds even strictly for $\gamma < 1$, and the second inequality follows from $\delta(N-1) < 1$. Hence, we can conclude that $p_M^* > p_R^*$, and consequently $q_M^* > q_R^*$ **Q.E.D.**

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